# .NET TECHNOLOGIES (01CE1523)
# Lab Manual

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**Class: 5EC1**
**Batch: A**

# Experiment 9

**AIM: Create Placement Dashboard using Data Controls.**

**Source: Database/Setup.sql**

```
USE master;
GO
IF DB_ID(N'PlacementPractical9') IS NULL
    CREATE DATABASE PlacementPractical9;
GO
USE PlacementPractical9;
GO
IF OBJECT_ID(N'dbo.Students', N'U') IS NULL
BEGIN
    CREATE TABLE dbo.Students (
        Id INT IDENTITY(1,1) PRIMARY KEY,
        Enrollment NVARCHAR(12) NOT NULL UNIQUE,
        StudentName NVARCHAR(80) NOT NULL,
        Department NVARCHAR(10) NOT NULL,
        Email NVARCHAR(100) NOT NULL,
        Company NVARCHAR(80) NULL,
        PackageLpa DECIMAL(6,2) NULL,
        Status NVARCHAR(10) NOT NULL,
        CONSTRAINT CK_Placement CHECK (
            (Status='Placed' AND Company IS NOT NULL AND
             PackageLpa IS NOT NULL AND PackageLpa>0) OR
            (Status='Pending' AND Company IS NULL AND PackageLpa IS NULL))
    );
    -- Fictional sample records for the lab demonstration.
    INSERT INTO dbo.Students
        (Enrollment,StudentName,Department,Email,Company,PackageLpa,Status)
    VALUES
    ('92400103460','Ayush Sharma','CE','ayush@example.com','TCS',7.00,'Placed'),
    ('92400103461','Ankit Patel','IT','ankit@example.com','Infosys',6.50,'Placed'),
    ('92400103462','Riya Shah','CE','riya@example.com','Accenture',12.50,'Placed'),
    ('92400103463','Rahul Mehta','EC','rahul@example.com',NULL,NULL,'Pending'),
    ('92400103464','Neha Joshi','IT','neha@example.com','Wipro',5.50,'Placed'),
    ('92400103465','Aman Verma','CE','aman@example.com',NULL,NULL,'Pending'),
    ('92400103466','Priya Desai','EC','priya@example.com','TCS',7.00,'Placed'),
    ('92400103467','Karan
Shah','IT','karan@example.com','Accenture',10.00,'Placed'),
    ('92400103468','Sneha Patel','CE','sneha@example.com','Infosys',6.50,'Placed'),
    ('92400103469','Dev Shah','EC','dev@example.com','Wipro',5.50,'Placed'),
    ('92400103470','Isha Mehta','IT','isha@example.com',NULL,NULL,'Pending'),
    ('92400103471','Jay Patel','CE','jay@example.com',NULL,NULL,'Pending');
END;
GO
```

**Source: Default.aspx**

```
<%@ Page Language="C#" AutoEventWireup="true" CodeBehind="Default.aspx.cs"
    Inherits="PlacementDashboard.Default" %>
<!DOCTYPE html>
<html lang="en">
<head runat="server">
    <meta charset="utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <title>Placement Dashboard</title>
    <link href="Content/Site.css" rel="stylesheet" />
</head>
<body><form id="form1" runat="server">
<header><div class="wrap">Placement Dashboard</div></header>
<main class="wrap">
    <h1>Campus Placement Dashboard</h1>
    <p>Student placement records | Sample dataset for lab demonstration</p>
    <div class="stats">
        <div class="stat">Total Students<strong><asp:Label ID="lblTotal"
runat="server" /></strong></div>
        <div class="stat">Placed<strong><asp:Label ID="lblPlaced" runat="server"
/></strong></div>
        <div class="stat">Placement Rate<strong><asp:Label ID="lblRate"
runat="server" /></strong></div>
        <div class="stat">Highest Package<strong><asp:Label ID="lblHighest"
runat="server" /> LPA</strong></div>
    </div>
    <p class="note">Summary figures cover all students. LPA = lakh per annum.</p>
    <section class="panel">
        <h2>Student Placement Records</h2>
        <div class="filters">
            <div><asp:Label runat="server" AssociatedControlID="ddlDepartment"
Text="Department" />
            <asp:DropDownList ID="ddlDepartment" runat="server" AutoPostBack="true"
                OnSelectedIndexChanged="FilterChanged">
                <asp:ListItem Value="All">All Departments</asp:ListItem>
                <asp:ListItem>CE</asp:ListItem><asp:ListItem>IT</asp:ListItem>
                <asp:ListItem>EC</asp:ListItem>
            </asp:DropDownList></div>
            <div><asp:Label runat="server" AssociatedControlID="ddlStatus"
Text="Placement Status" />
            <asp:DropDownList ID="ddlStatus" runat="server" AutoPostBack="true"
                OnSelectedIndexChanged="FilterChanged">
                <asp:ListItem Value="All">All Statuses</asp:ListItem>

<asp:ListItem>Placed</asp:ListItem><asp:ListItem>Pending</asp:ListItem>
            </asp:DropDownList></div>
        </div>
```

**Source: Default.aspx (continued)**

```
        <div class="table-wrap">
        <asp:GridView ID="gvStudents" runat="server" DataSourceID="dsStudents"
            AutoGenerateColumns="false" DataKeyNames="Id" CssClass="grid"
            AllowPaging="true" PageSize="6" AllowSorting="true"
            OnPageIndexChanging="GridChanging" OnSorting="GridSorting"
            EmptyDataText="No placement records match these filters."
            OnSelectedIndexChanged="StudentSelected">
            <Columns>
                <asp:BoundField DataField="Enrollment" HeaderText="Enrollment" />
                <asp:BoundField DataField="StudentName" HeaderText="Student Name"
                    SortExpression="StudentName" />
                <asp:BoundField DataField="Department" HeaderText="Dept."
                    SortExpression="Department" />
                <asp:BoundField DataField="Company" HeaderText="Company"
                    NullDisplayText="-" />
                <asp:BoundField DataField="PackageLpa" HeaderText="Package (LPA)"
                    DataFormatString="{0:0.00}" NullDisplayText="-"
                    SortExpression="PackageLpa" />
                <asp:BoundField DataField="Status" HeaderText="Status" />
                <asp:CommandField ShowSelectButton="true" SelectText="View" />
            </Columns>
            <SelectedRowStyle CssClass="selected" />
            <PagerStyle CssClass="pager" />
        </asp:GridView></div>
    </section>
    <section class="panel">
        <h2>Selected Student Details</h2>
        <asp:DetailsView ID="dvStudent" runat="server" DataSourceID="dsDetails"
            AutoGenerateRows="false" CssClass="details"
            EmptyDataText="Select View beside a student to see full details.">
            <Fields>
                <asp:BoundField DataField="Enrollment" HeaderText="Enrollment" />
                <asp:BoundField DataField="StudentName" HeaderText="Student Name" />
                <asp:BoundField DataField="Department" HeaderText="Department" />
                <asp:BoundField DataField="Email" HeaderText="Email" />
                <asp:BoundField DataField="Company" HeaderText="Company"
NullDisplayText="-" />
                <asp:BoundField DataField="PackageLpa" HeaderText="Package (LPA)"
                    DataFormatString="{0:0.00}" NullDisplayText="-" />
                <asp:BoundField DataField="Status" HeaderText="Status" />
            </Fields>
        </asp:DetailsView>
    </section>
    <section class="panel">
        <h2>Company Summary</h2>
        <p class="note">All placed students, grouped by company.</p>
```

**Source: Default.aspx (continued)**

```
        <div class="companies">
        <asp:Repeater ID="rptCompanies" runat="server" DataSourceID="dsCcompanies">
            <ItemTemplate>
                <article class="company"><strong><%#: Eval("Company") %></strong>
                <p>Students placed: <%#: Eval("PlacedCount") %></p>
                <p>Highest: <%#: Eval("HighestPackage", "{0:0.00}") %>
LPA</p></article>
            </ItemTemplate>
        </asp:Repeater></div>
    </section>
</main>
<footer class="wrap">Practical 9 | ASP.NET Data Controls<br />
    AYUSH SHARMA | 92400103460 | 5EC1 | Batch A</footer>
<asp:SqlDataSource ID="dsStudents" runat="server"
    ConnectionString="<%$ ConnectionStrings:PlacementDb %>"
    SelectCommand="SELECT
Id,Enrollment,StudentName,Department,Company,PackageLpa,Status
        FROM Students WHERE (@Department='All' OR Department=@Department)
        AND (@Status='All' OR Status=@Status) ORDER BY Id">
    <SelectParameters>
        <asp:ControlParameter Name="Department" ControlID="ddlDepartment"
            PropertyName="SelectedValue" Type="String" DefaultValue="All" />
        <asp:ControlParameter Name="Status" ControlID="ddlStatus"
            PropertyName="SelectedValue" Type="String" DefaultValue="All" />
    </SelectParameters>
</asp:SqlDataSource>
<asp:SqlDataSource ID="dsDetails" runat="server"
    ConnectionString="<%$ ConnectionStrings:PlacementDb %>"
    SelectCommand="SELECT
Enrollment,StudentName,Department,Email,Company,PackageLpa,Status
        FROM Students WHERE Id=@Id">
    <SelectParameters>
        <asp:ControlParameter Name="Id" ControlID="gvStudents"
            PropertyName="SelectedValue" Type="Int32" DefaultValue="0" />
    </SelectParameters>
</asp:SqlDataSource>
<asp:SqlDataSource ID="dsCompanies" runat="server"
    ConnectionString="<%$ ConnectionStrings:PlacementDb %>"
    SelectCommand="SELECT Company,COUNT(*) AS PlacedCount,MAX(PackageLpa) AS
HighestPackage
        FROM Students WHERE Status='Placed' GROUP BY Company ORDER BY Company" />
</form></body></html>
```

**Source: Default.aspx.cs**

```
using System;
using System.Configuration;
using System.Data.SqlClient;
using System.Web.UI;
using System.Web.UI.WebControls;

namespace PlacementDashboard
{
    public partial class Default : Page
    {
        protected void Page_Init(object sender, EventArgs e)
        {
            // Bind ViewState to this browser session.
            Session["DashboardStarted"] = true;
            ViewStateUserKey = Session.SessionID;
        }
        protected void Page_Load(object sender, EventArgs e)
        {
            if (!IsPostBack) LoadSummary();
        }
        private void LoadSummary()
        {
            string cs =
ConfigurationManager.ConnectionStrings["PlacementDb"].ConnectionString;
            using (var connection = new SqlConnection(cs))
            using (var command = new SqlCommand(
                "SELECT COUNT(*),COALESCE(SUM(CASE WHEN Status='Placed' THEN 1 ELSE
0 END),0)," +
                "COALESCE(MAX(PackageLpa),0) FROM Students", connection))
            {
                connection.Open();
                using (var reader = command.ExecuteReader())
                {
                    reader.Read();
                    int total = reader.GetInt32(0), placed = reader.GetInt32(1);
                    lblTotal.Text = total.ToString();
                    lblPlaced.Text = placed.ToString();
                    lblRate.Text = (total == 0 ? 0 : placed * 100.0 /
total).ToString("0.0") + "%";
                    lblHighest.Text = reader.GetDecimal(2).ToString("0.00");
                }
            }
        }
        private void ClearSelection()
        {
            gvStudents.SelectedIndex = -1;
```

**Source: Default.aspx.cs (continued)**

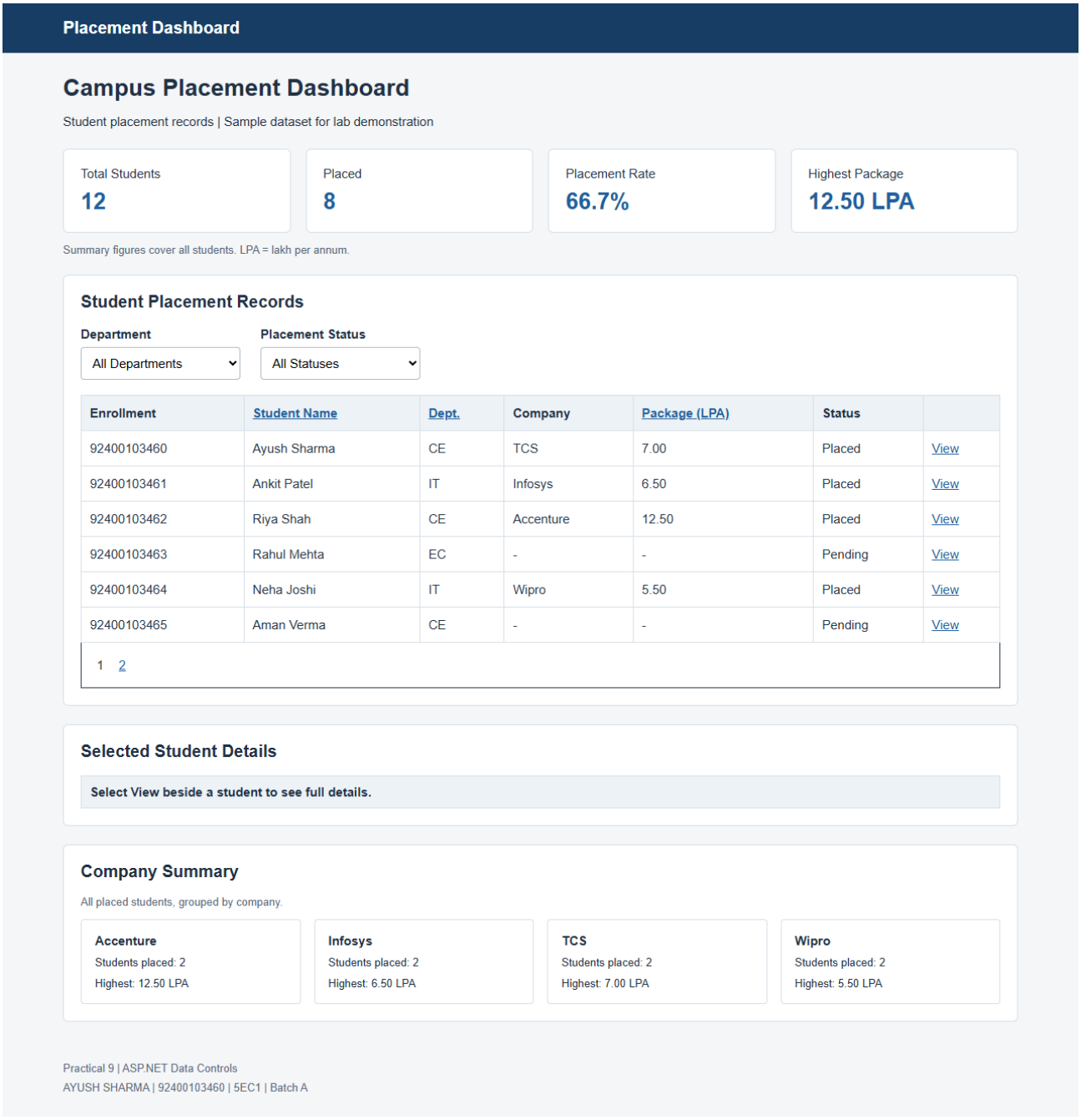
```
            dvStudent.DataBind();
        }
        protected void FilterChanged(object sender, EventArgs e)
        {
            gvStudents.PageIndex = 0;
            ClearSelection();
            gvStudents.DataBind();
        }
        protected void GridChanging(object sender, GridViewPageEventArgs e)
        {
            ClearSelection();
        }
        protected void GridSorting(object sender, GridViewSortEventArgs e)
        {
            ClearSelection();
        }
        protected void StudentSelected(object sender, EventArgs e)
        {
            dvStudent.DataBind();
        }
    }
}
```

**Source: Web.config**

```
<?xml version="1.0"?>
<configuration>
  <connectionStrings>
    <add name="PlacementDb" providerName="System.Data.SqlClient"
      connectionString="Data Source=(LocalDB)\MSSQLLocalDB;Initial
Catalog=PlacementPractical9;Integrated Security=True;" />
  </connectionStrings>
  <system.web>
    <compilation debug="true" targetFramework="4.8" />
    <httpRuntime targetFramework="4.8" />
    <pages controlRenderingCompatibilityVersion="4.0" />
  </system.web>
</configuration>
```

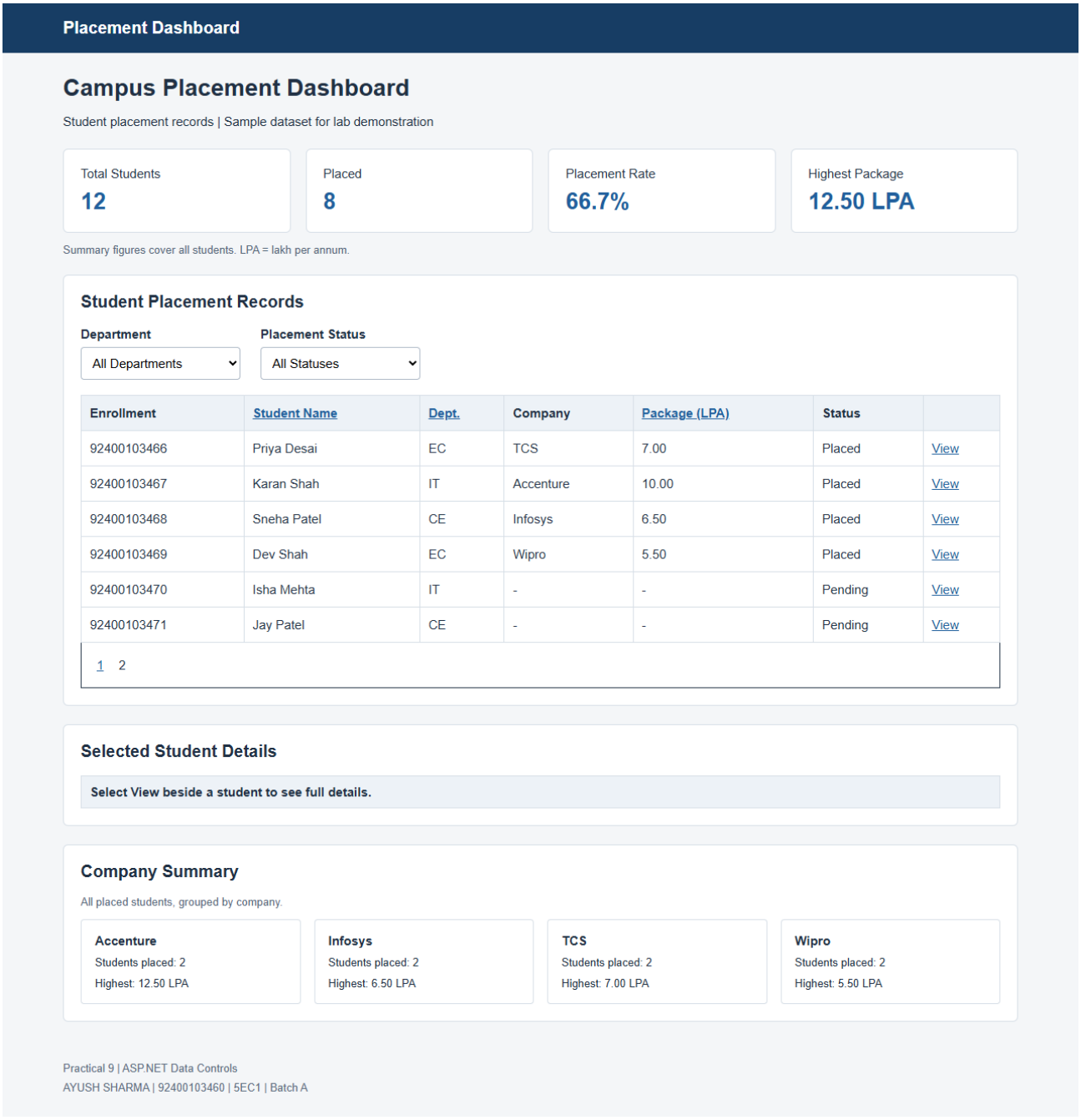
## Output: Screenshot:

### Dashboard: Summary labels, GridView and company Repeater



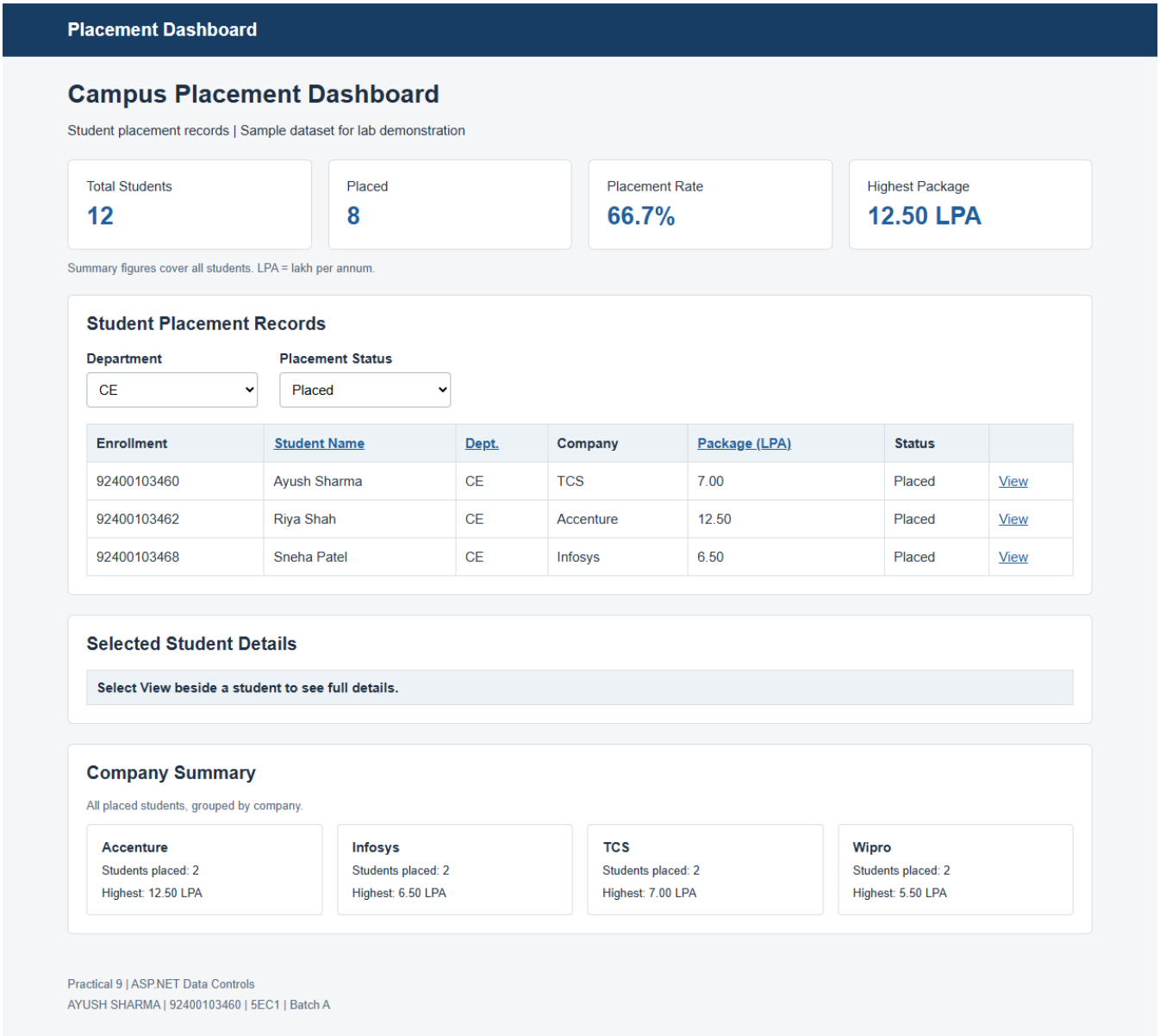
## Output: Screenshot:

### GridView: Second page of placement records



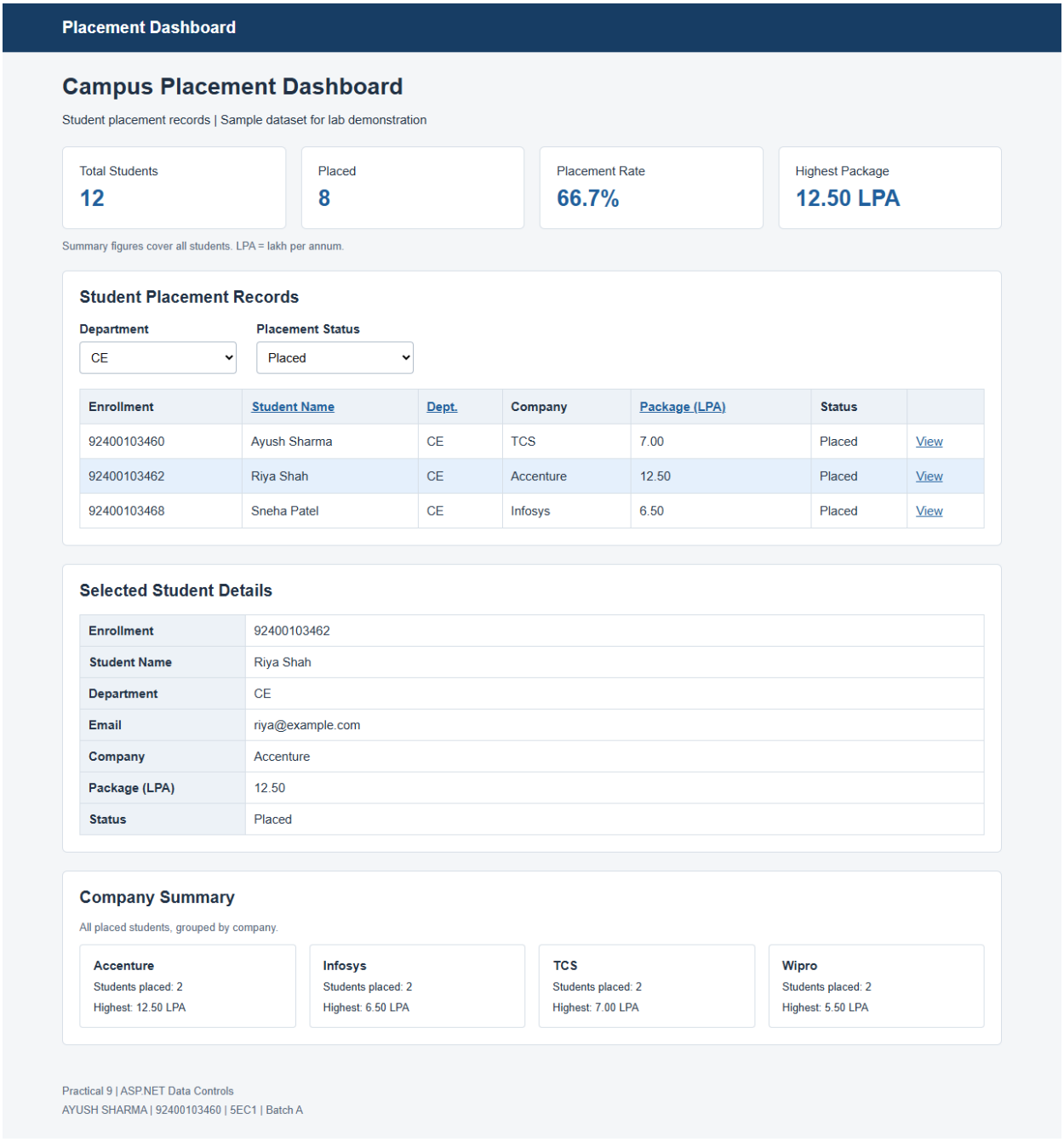
## Output: Screenshot:

### DropDownList: Combined department and placement-status filters



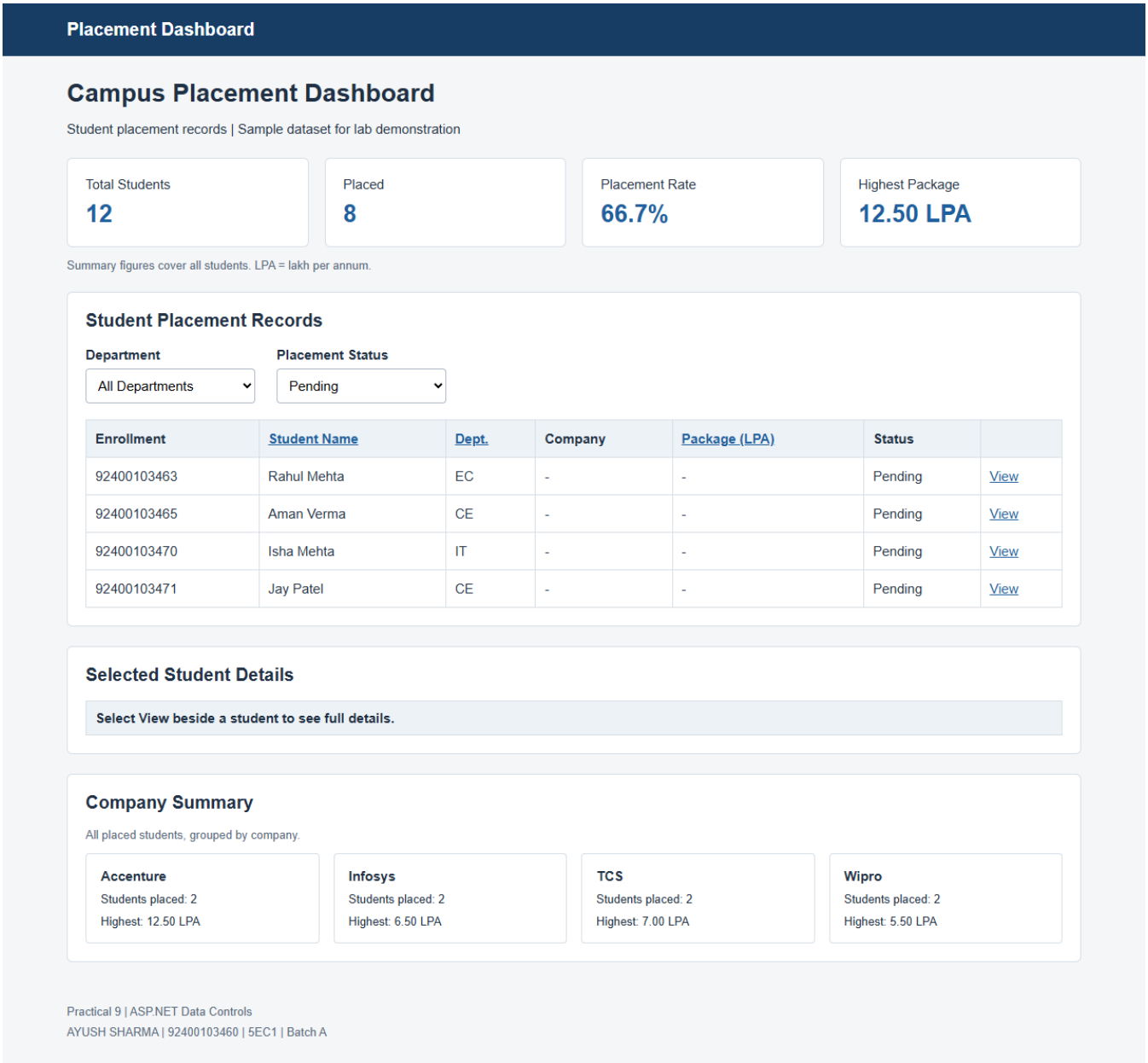
## Output: Screenshot:

### DetailsView: Selected student placement information



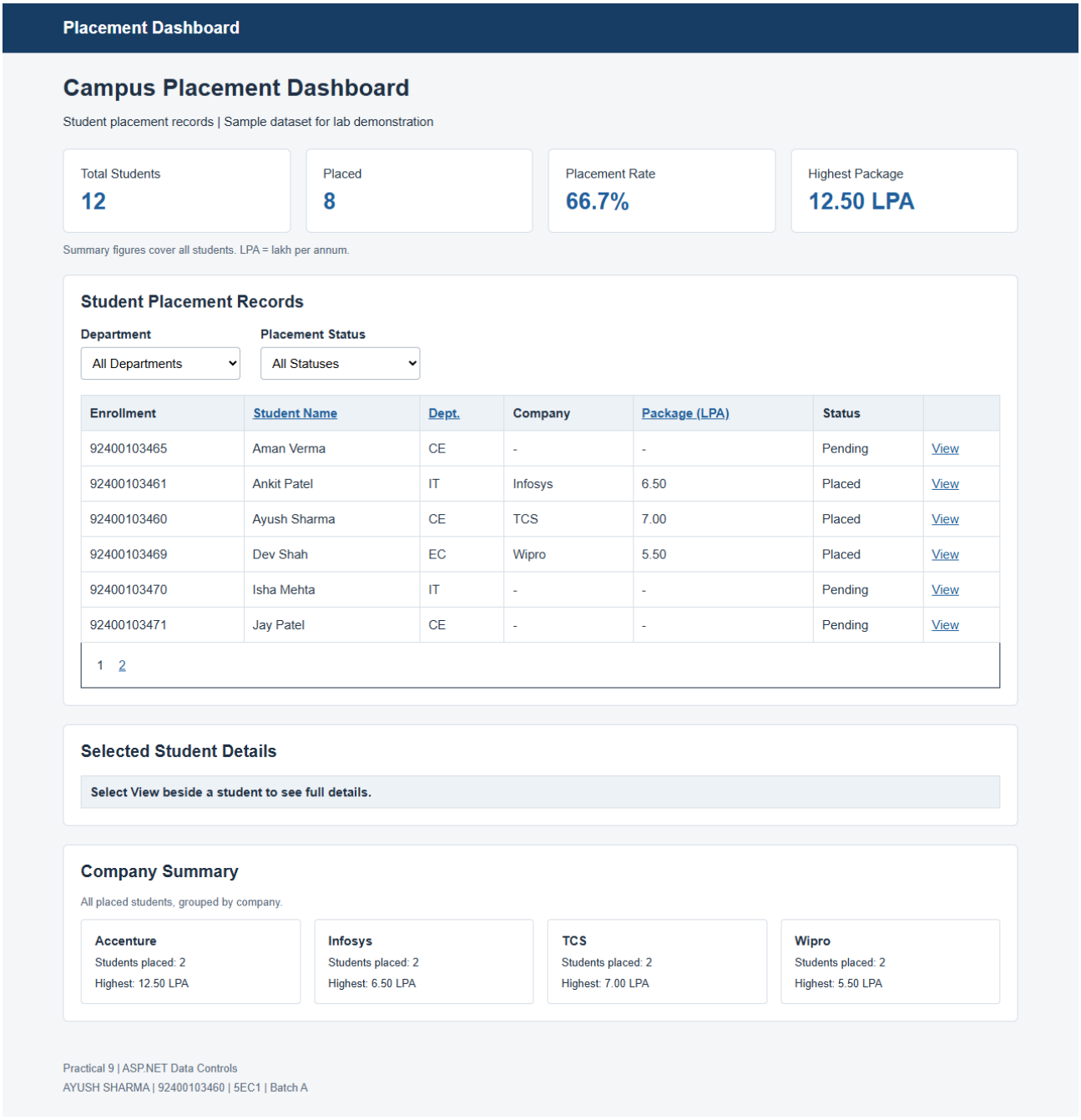
## Output: Screenshot:

### GridView: Students with pending placement status



## Output: Screenshot:

### GridView: Student records sorted by name




## GitHub URL:

https://github.com/ius-sharma/Placement-Dashboard